A 4×1 Microstrip Patch Antenna Array with Stacked-Patch Bandwidth Enhancement

Jack Basinski
EE 464/574, University of Washington, Seattle
jbasin@uw.edu

***Abstract*—This report discusses the design, simulation, fabrication, and validation of a $5.8\,\mathrm{GHz}$ 4×1 microstrip patch antenna array on a $1.6\,\mathrm{mm}$ FR-4 substrate ($\varepsilon_r = 4.4$, $\tan\delta = 0.02$). A corporate feeding network with two quarter-wave transformers was implemented to impedance-match each antenna element to the $50\,\Omega$ SMA input port. Simulation results demonstrate a bandwidth exceeding $100\,\mathrm{MHz}$ at VSWR ≤ 2. Additionally, a broadside gain above $8\,\mathrm{dBi}$, a fan-beam beamwidth narrower than $25^\circ\times100^\circ$, and linear polarization was achieved. To enhance bandwidth, a stacked patch array configuration was implemented. The stacked design was centered at 5.5 GHz and achieved a bandwidth of 1.3 GHZ while maintaining the gain and radiation characteristics of the 4×1 array. Both configurations met or exceeded all design specifications.**

I. Introduction

Microstrip patch antennas are essential to modern communication systems because they are low-cost and can be easily fabricated on PCBs [1]. By modifying the antenna array elements, the radiation pattern can be customized to achieve numerous objectives.

The Objective of this project is to design, simulate, fabricate, and measure a 4×1 array of microstrip patch antennas, resonant at $5.8\,\mathrm{GHz}$, fed by a corporate feeding network. The required specifications are summarized in Table I. Section II discusses the microstrip antenna and feeding network design procedure. Section III reports on simulation and measurement results. Section IV builds on the original design with a stacked parasitic patch configuration, demonstrating bandwidth enhancement from 258 MHz to over 1.3 GHz. Section V concludes and discusses future work.

TABLE I: Required Design Specifications

Parameter	Requirement
Centre frequency	$5.8\,\mathrm{GHz}$
Bandwidth	$\geq 100\,\mathrm{MHz}$ at VSWR ≤ 2
VSWR	≤ 2
Gain	$\geq 8\,\mathrm{dBi}$
Beamwidth	$< 25^\circ \times 100^\circ$
Polarization	Linear (vertical)

II. Theory

A. Microstrip Patch Antenna

Microstrip patch antennas, as described by Balanis (Ch. 14) [1], consist of a thin metallic patch of width W_p and length L_p, sitting on a grounded substrate of thickness h with relative permittivity ε_r. The patch primarily radiates through its two open edges, and is modelled as two parallel radiating slots separated by L_p.

Patch Width:

$$W_p = \frac{c}{2f_r}\sqrt{\frac{2}{\varepsilon_r+1}}, \tag{1}$$

where c is the speed of light and f_r the resonant frequency.

Effective Permittivity:

$$\varepsilon_{r,\mathrm{eff}} = \frac{\varepsilon_r+1}{2} + \frac{\varepsilon_r-1}{2}\left(1+\frac{12h}{W_p}\right)^{-1/2}. \tag{2}$$

Fringing-Field Extension and Physical Length:

$$\Delta L = 0.412\,h\,\frac{(\varepsilon_{r,\mathrm{eff}}+0.3)(W_p/h+0.264)}{(\varepsilon_{r,\mathrm{eff}}-0.258)(W_p/h+0.8)}, \tag{3}$$

$$L_p = \frac{c}{2f_r\sqrt{\varepsilon_{r,\mathrm{eff}}}} - 2\Delta L. \tag{4}$$

Edge Radiation Resistance: From the transmission-line model, the radiation conductance per slot is [1]:

$$G_1 = \frac{1}{90}\left(\frac{W_p}{\lambda_0}\right)^2, \tag{5}$$

for $h \ll \lambda_0$ this gives an imput resistance at resonance of:

$$R_{\mathrm{in}} = \frac{1}{2(G_1 \pm G_{12})} \approx 90\left(\frac{\varepsilon_r^2}{\varepsilon_r-1}\right)\left(\frac{L_p}{W_p}\right)^2, \tag{6}$$

where G_{12} is the mutual conductance between slots. For an edge-fed patch, $R_{\mathrm{in}} \approx 100$–$300\ \Omega$ depending on W_p and ε_r.

B. Quarter-Wave Transformer Matching

A quarter-wave transformer (QWT) of characteristic impedance $Z_c^{(n)}$ transforms a load Z_L to [2]:

$$Z_{\mathrm{in}} = \frac{\left(Z_c^{(n)}\right)^2}{Z_L} \tag{7}$$

For the two-stage corporate feed, the impedance chain from the antenna edge resistance R_{in} to the $50\,\Omega$ port is:

a) Stage 1 — QWT$_2$ (antenna to first T-junction)::

$$Z_c^{(2)} = \sqrt{Z_c^{(1)}\cdot R_{\mathrm{in}}}. \tag{8}$$

b) Stage 2 — QWT$_1$ (T-junction to main line)::

$$Z_c^{(1)} = \sqrt{Z_0\cdot\frac{Z_c^{(2)}}{2}}, \tag{9}$$

where $Z_0 = 50\,\Omega$.

Microstrip Line Synthesis: The closed-form synthesis equations of Pozar (§3.8) [2] give, for $W/h \leq 1$:

$$Z_c = \frac{60}{\sqrt{\varepsilon_{r,\text{eff}}}} \ln\left(\frac{8h}{W} + \frac{W}{4h}\right), \tag{10}$$

and for $W/h \geq 1$:

$$Z_c = \frac{120\pi}{\sqrt{\varepsilon_{r,\text{eff}}}\,[W/h + 1.393 + 0.667\ln(W/h + 1.444)]}. \tag{11}$$

The physical QWT length is set by the guided wavelength:

$$\lambda_g = \frac{c}{f_r\sqrt{\varepsilon_{r,\text{eff}}}}, \qquad L_{\text{QWT}} = \frac{\lambda_g}{4}. \tag{12}$$

C. Single Patch Element

The free-space wavelength at 5.8 GHz is:

$$\lambda_0 = \frac{c}{f} = \frac{3\times 10^8}{5.8\times 10^9} \approx 51.7\,\text{mm}. \tag{13}$$

Applying (1)–(4) with $\varepsilon_r = 4.4$, $h = 1.6\,\text{mm}$, and $f_r = 5.8\,\text{GHz}$ yields the initial patch dimensions in Table II.

TABLE II: Single Patch Design Parameters

Parameter	Equation	Value (mm)
W_p	(1)	10.9
$\varepsilon_{r,\text{eff}}$	(2)	3.84
ΔL	(3)	0.723
L_p	(4)	13.19
R_{in} (Ω)	(6)	285.6

D. Corporate Feeding Network

The fixed narrow feed line ($W_4 = 0.25\,\text{mm}$) connects the edge of each patch element while acting as the first impedance transformer. Its characteristic impedance on 1.6 mm FR-4, computed from the synthesis equations, is approximately 138.8 Ω. The full feeding network parameters are given in Table III.

TABLE III: Feeding Network Design Parameters

Param.	Value	Param.	Value
W_0 (mm)	6.08	L_1 (mm)	10
W_1 (mm)	0.709	L_2 (mm)	7.158
W_2 (mm)	2.36	L_3 (mm)	13.7
W_3 (mm)	1.78	$L_{4(a,b)}$ (mm)	3.8
W_4 (mm)	0.25	D	$0.68\lambda_0$
λ_0 (mm)	51.724	L_{PCB} (mm)	160
W_{PCB} (mm)	80		

E. HFSS Model Setup

The antenna array was modelled using Ansys HFSS. The dielectric layer was set to FR-4 substrate ($\varepsilon_r = 4.4$, $\tan\delta = 0.02$, $h = 1.6\,\text{mm}$, $t_{\text{Cu}} = 18\,\mu\text{m}$), and a wave port SMA input. The SMA coaxial connector was composed of a copper core (R= 1.27 mm) and Teflon dielectric (R= 4.11 mm, $\varepsilon_r = 2.1$, $\tan\delta = 0.001$). A radiation boundary was set $\lambda/4$ from all radiating surfaces, and a frequency sweep from 4 GHz to 7 GHz was used for S_{11} and impedance measurements. Fig. 1 shows the general design of the microstrip patch antenna array and feeding network. Fig. 2 shows simulated radiation field overlays, demonstrating the feeding network's capabilities and the fan radiation pattern.

After testing and optimization in HFSS, antenna traces were exported, and Altium Designer was used to design the antennas for fabrication.

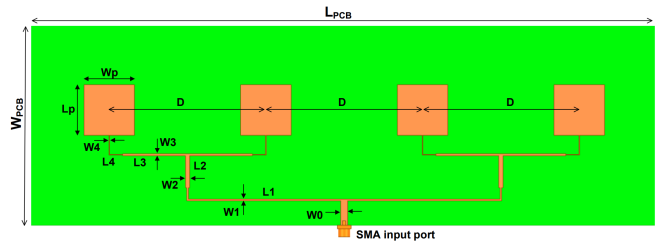

Fig. 1: Microstrip Antenna Array Schematic

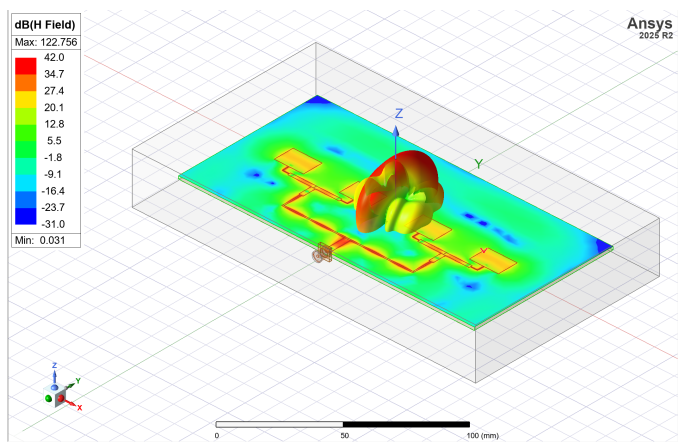

Fig. 2: Antenna Field Overlays

III. Simulation and Measurement Results

A. Input Match

Fig. 3 and 4 show the simulated and measured S_{11} response respectively. The simulated resonance occurs at 5.79 GHz with a bandwidth of 258 MHz, satisfying the 100 MHz specification. The measured resonance occurs at 5.76 GHz with a bandwidth of 632 MHZ. Looking at Fig. 5, the real and imaginary components of Z_{in} at resonance are 32.56 Ω, and 8.05 Ω respectively. These values are consistent across the antenna's bandwidth, confirming good matching to the 50 Ω port.

B. Radiation Pattern and Gain

Fig. 6 shows the 3-D gain pattern at 5.8 GHz. The maximum gain is 9.18 dBi, exceeding the $\geq 8\,\text{dBi}$ requirement. Fig. 7 presents the E- and H-plane cuts. The half-power beamwidths are 127.84° (E-plane) and 20.87° (H-plane). Fig. 8 shows the antenna's measured radiation pattern in the E and H planes.

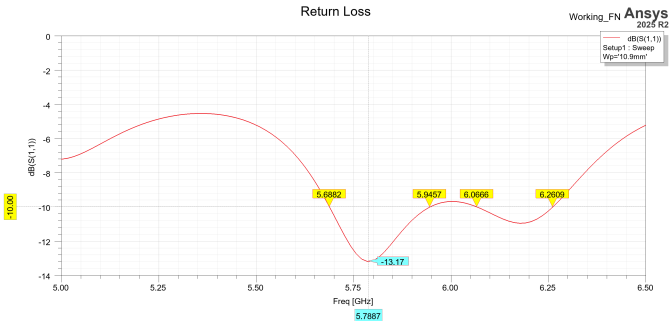

Fig. 3: Simulated S_{11} vs. frequency.

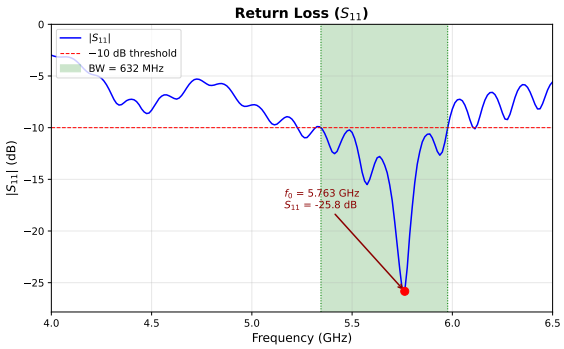

Fig. 4: Measured S_{11} vs. frequency.

The half-power beamwidths are $114.7°$ (E-plane) and $22.2°$ (H-plane) respectively.

C. Antenna Efficiency

The simulated radiation efficiency at $5.8\,\mathrm{GHz}$ is $\eta_{\mathrm{rad}} = 56.1\%$. Losses are attributable to the dielectric loss tangent ($\tan\delta = 0.02$) of FR-4 and the finite copper thickness ($t = 18\,\mu\mathrm{m}$).

D. Axial Ratio and Polarization

Fig. 9 shows the axial ratio in the E- and H-planes. The axial ratio at $\theta = 0°$ is approximately 77dB, and is greater than 25dB in all of the primary lobe, confirming linear vertical polarization.

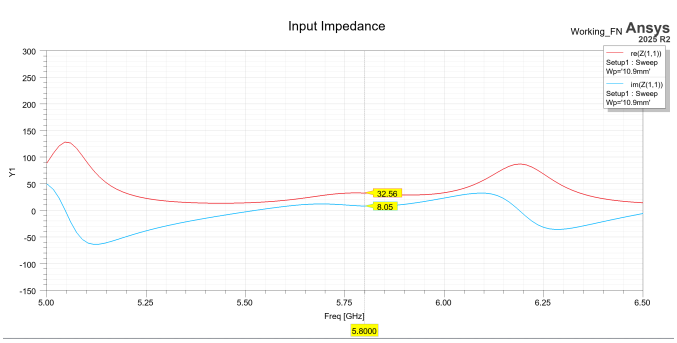

Fig. 5: Simulated input impedance Z_{in} vs. frequency.

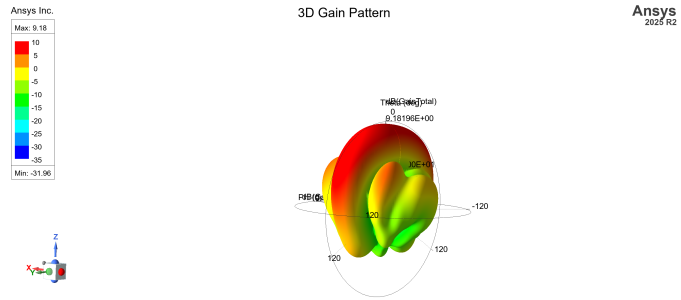

Fig. 6: Simulated 3-D gain pattern at $5.8\,\mathrm{GHz}$.

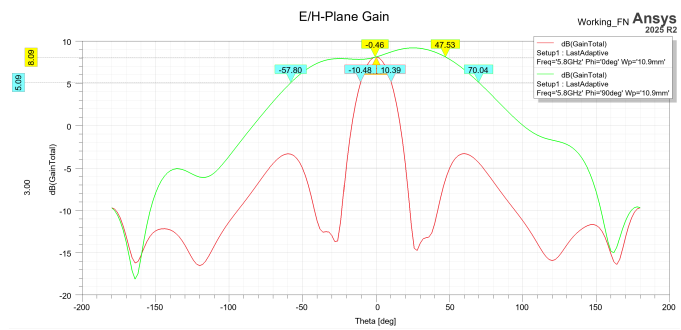

Fig. 7: Simulated gain pattern in E- and H-plane.

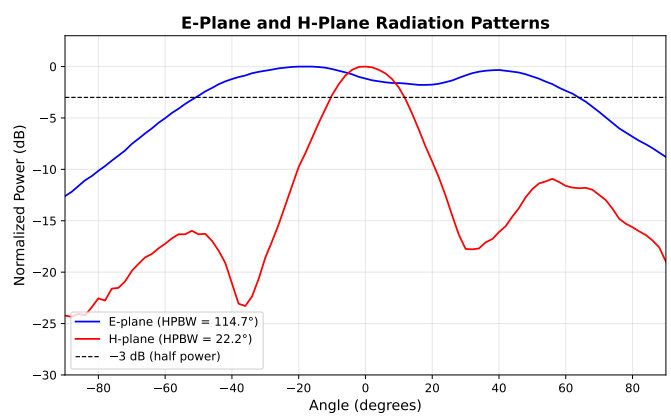

Fig. 8: Measured gain pattern in E- and H-plane, normalized.

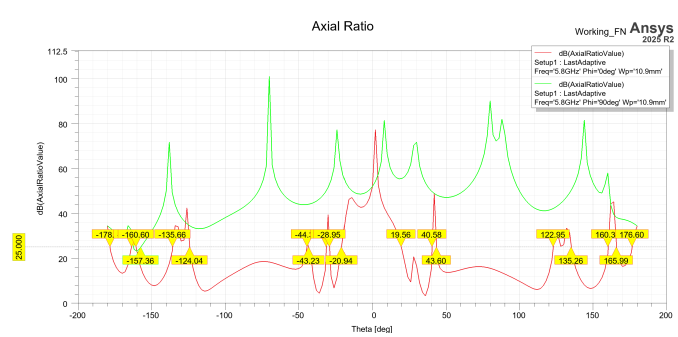

Fig. 9: Simulated axial ratio in E-/H-planes at $5.8\,\mathrm{GHz}$.

E. Simulation vs. Measurement Comparison

Table IV summarises all simulated and measured metrics. The fabricated antenna performed better than expected: The measured return loss was 10 dB lower than the simulation results, and the bandwidth was twice as large. Differences in expected performance can be attributed to variation in the connection strength between the SMA port and the coaxial cable, as well as between the SMA port and the solder joint.

The simulated and measured radiation patterns of the antenna are extremely similar to one another, with the half-power beamwidths only differing by <10°. Variation in the half-power beamwidth of the E-plane can be attributed to radiation interference from the feeding network. This can be remedied by minimizing radiating components in the network such as T-junctions. Although the E plane does not perfectly satisfy the $25° \times 100°$ specification, it is not the dominant coverage plane, so the variance is acceptable. The simulated radiation patterns are highly accurate to the measured patterns because they depend on patch dimensions and element spacing; factors that are highly controllable in PCB fabrication.

TABLE IV: Simulated vs. Measured Performance Summary

Parameter	Spec	Sim.	Meas.
Centre frequency (GHz)	5.8	5.79	5.76
Bandwidth (MHz), VSWR ≤ 2	≥ 100	258	632
Peak gain (dBi)	≥ 8	9.18	N/A
E-plane HPBW (°)	< 100	127.84	114.7
H-plane HPBW (°)	< 25	20.87	22.2
Axial ratio at 0° (dB)	> 3	77 dB	N/A
Radiation efficiency (%)	N/A	56.1	N/A

IV. Bandwidth Enhancement via Stacked Parasitic Patch

A. Motivation

The Baseline microstrip antenna array previously discussed achieves all design objectives and is appropriate for applications such as GPS and radio services. However, microstrip antenna arrays are inherently narrowband [3], and further bandwidth enhancement is desirable for higher data rates or tolerance to manufacturing or environmental variations. For the current antenna design, a simple remedy is a stacked-patch configuration, in which a parasitic patch array is placed above the driven patch. This is a well-established technique for increasing bandwidth without significantly increasing the footprint of the antenna [4].

B. Coupled-Resonator Framework

Like the microstrip patch antenna, a stacked patch antenna is modelled as two coupled resonators. When the parasitic patch array is in close proximity to the driven array, the two systems electromagnetically couple and exhibit two resonances. For coupling coefficient k and nominal center frequency f_0, the two resonant frequencies are approximately:

$$f_1 \approx f_0(1 - \frac{k}{2}), \qquad f_2 \approx f_0(1 + \frac{k}{2}). \tag{14}$$

where the frequency separation Δf is:

$$\Delta f = f_2 - f_1 = 2kf_0 \tag{15}$$

The two resonances merge to form a single broadened impedance match. The bandwidth enhancement factor is 1.5–2× that of a single-patch design.

C. Design Approach

The stacked configuration was designed with a center frequency of $f_0 = 5.5\,\mathrm{GHz}$ to ensure the upper resonance remained within the measurement system's $6.5\,\mathrm{GHz}$ limit. Initial characterization of a single driven patch at the center frequency yielded a baseline bandwidth of approximately $110\,\mathrm{MHz}$. A coupling coefficient of $k = 0.04$ was selected, and the target resonant frequencies for each array were calculated to be:

$$f_1 = 5.5 \times (1 - 0.02) = 5.39\,\mathrm{GHz}, \tag{16}$$

$$f_2 = 5.5 \times (1 + 0.02) = 5.61\,\mathrm{GHz}. \tag{17}$$

With a frequency separation of approximately $220\,\mathrm{MHz}$. The merged response was expected to yield a combined bandwidth on the order of $1.5\times$ the baseline ($\geq 165\,\mathrm{MHz}$).

Parasitic Patch Sizing: The parasitic patch array was designed larger than the driven array so that it would resonate at the lower frequency (f_1). Likewise, the driven array was re-designed to resonate at f_2. These initial patch decisions were calculated simultaneously with standard transmission line equations (1)–(4), for reasons described by Pozar and Duffy [4]: "Due to strong coupling between the patches, the resonant sizes of the patches must be determined simultaneously." After the initial design was completed, the patches were individually tuned using a trial-and-error approach.

Air-Gap Spacer: A $5\,\mathrm{mm}$ air gap was standardized for this implementation using brass standoffs at each corner of the driven patch. $5\,\mathrm{mm}$ is approximately $\frac{1}{10}\lambda_0$, corresponding to moderate coupling strength and a broadened bandwidth where good impedance matching is still achievable. Fig. 10 shows an isometric view of the final stacked patch array system.

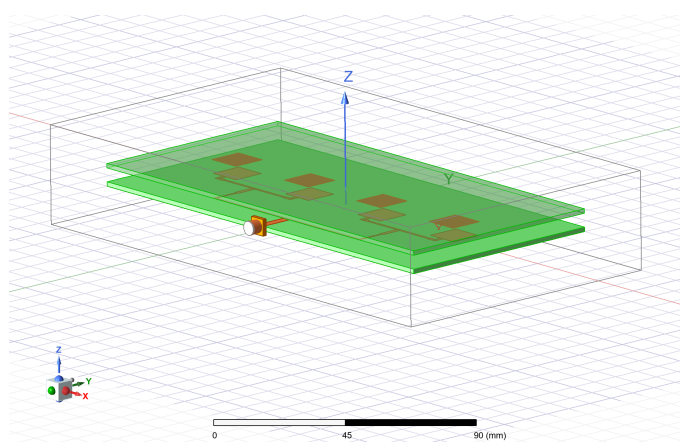

Fig. 10: Final stacked patch array system

Optimization: The stacked parasitic patch arrays have numerous interacting parameters, but are primarily governed by driven patch dimensions, parasitic patch dimensions, and feed network geometry. This makes design by purely calculation impractical. Instead, the calculated patch dimensions were used as initial values in HFSS, and the design was iteratively refined by observing the S_{11} response. The optimization targeted:

1) Two resonant dips in the S_{11} response near f_1 and f_2,
2) Sufficient overlap of the resonances corresponding to one continuous bandwidth,
3) Unchanged behavior in the system's radiation pattern and gain.

D. Stacked Array Configuration

The final configuration for the stacked array saw a redesigned feeding network and patch dimensions on the driven patch. The parasitic layer was designed without a ground plane or feeding network. Table V summarizes the key design parameters.

TABLE V: Stacked-Patch Design Parameters

Parameter	Value
Target center frequency, f_0	5.5 GHz
Coupling coefficient, k	0.04
Lower resonance, f_1	5.39 GHz
Upper resonance, f_2	5.61 GHz
Air-gap height	5 mm
Driven patch dimensions	11.9mm $\times$ 11.9mm
Parasitic patch dimensions	12.61 $\times$ 12.61mm
Baseline single-patch bandwidth	110 MHz
Target enhanced bandwidth	$\geq$165 MHz

E. Results

Fig. 11 and Fig.12 show the simulated and measured S_{11} response of the stacked-patch array, respectively. Both have resonances approximately at 5.3 GHz and 5.61 GHz. The simulated bandwidth is 1.9 GHz, and the measured bandwidth is 1.3 GHz. Both results show two closely spaced minima combining to form a wide impedance match.

Fig. 13 shows the simulated input impedance of the system. There are three resonances where the imaginary component of the input resistance crosses zero. At each resonance, the real component is approximately 50 Ω, indicating a strong match across the entire bandwidth.

Fig. 14 and Fig.15 show the simulated and measured radiation pattern of the stacked-patch array. The simulated maximum gain is 11.1 dBi, and the corresponding half-power beamwidths are 66.1° (E-plane) and 20.51° (H-plane). The measured results reported beamwidths of 54.2° (E-plane) and 18.9° (H-plane). These results maintain the same directional behavior of the original 4×1 patch array. Additionally, Fig. 16 shows the stacked patch array in action as the H-Field couples with the parasitic patches.

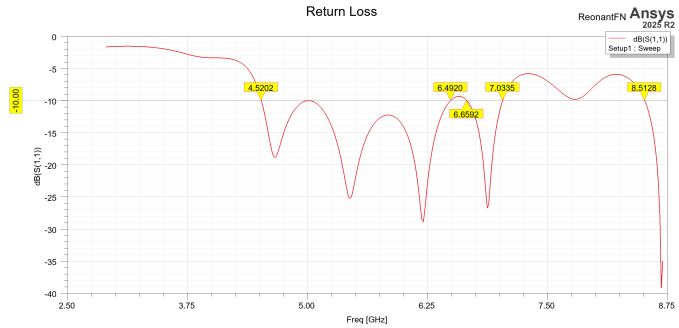

Fig. 11: Stacked patch array simulated return loss

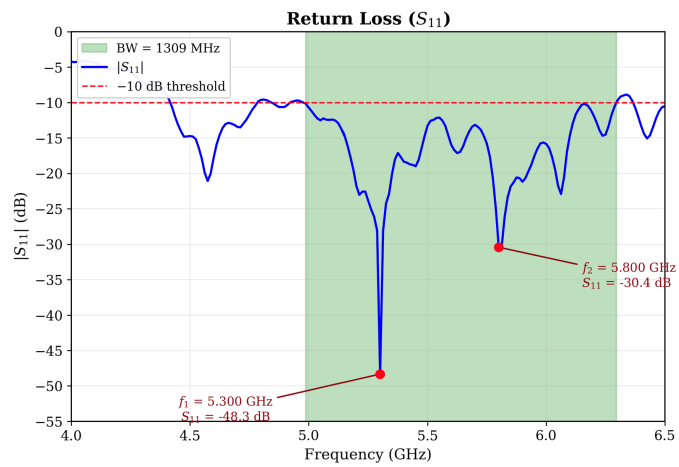

Fig. 12: Stacked patch array measured return loss

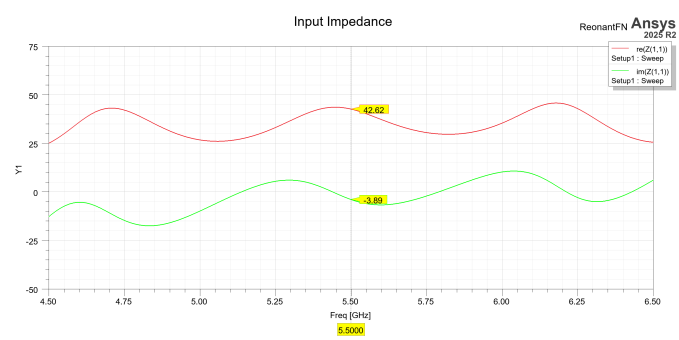

Fig. 13: Stacked patch array simulated input impedance

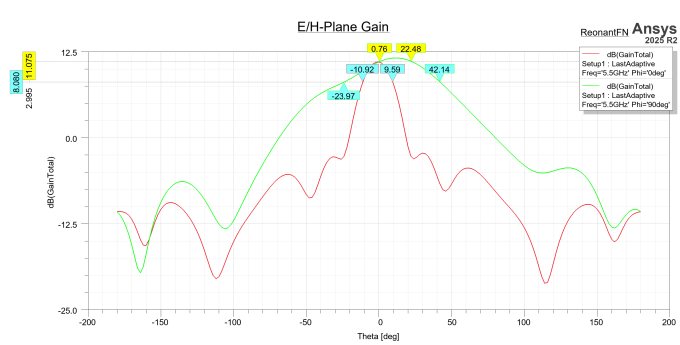

Fig. 14: Simulated radiation pattern in E- and H-plane

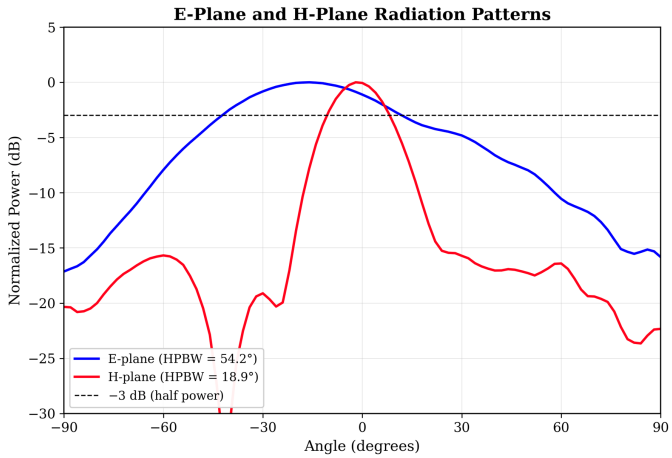

Fig. 15: Measured gain pattern of E- and H-plane, normalized

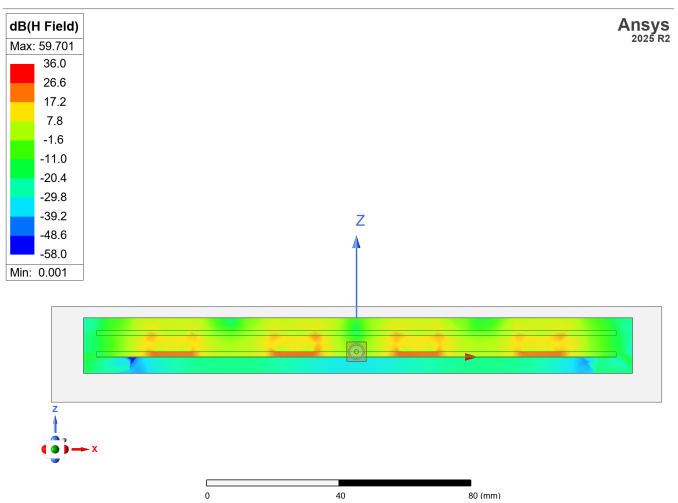

Fig. 16: H-Field overlay on stacked patch array

Table VI summarizes and compares the baseline and stacked-patch array performance.

TABLE VI: Baseline vs. Stacked-Patch Performance

Parameter	Baseline	Stacked
Center frequency (GHz)	5.71	5.5
Bandwidth (MHz), VSWR ≤ 2	258	1309
Peak gain (dBi)	9.18	11.1
Radiation efficiency (%)	56.1	63.5

F. Discussion

The stacked-patch approach successfully demonstrated the coupled-resonance bandwidth enhancement effect. Several observations merit discussion:

- **Resonance inconsistencies** S_{11} plots report two distinct resonances at approximately 6.25 GHz and 7 GHz. The exact cause is uncertain, but it is likely due to higher-order radiating modes of the patches. These resonances are outside the intended operating band and do not significantly impact the primary design objectives.
- **Resonance Shifting** The upper resonance of the system was measured at 5.8 GHz, approximately 200 MHz higher than the simulated value of 5.61 GHZ. This shift is attributed to an air gap slightly smaller than designed, thereby increasing the effective coupling coefficient. The stronger coupling, combined with higher-order resonances, produced a significantly wider bandwidth than initially predicted.
- **Gain considerations:** The simulated radiation pattern of the stacked array system remains consistent with the original design. However, the measured pattern shows significant distortion to the H-plane. This can be attributed to misalignment of the parasitic array.

This extension demonstrates that bandwidth enhancement is achievable with low complexity and advertises the stacked-patch configuration as an appealing option for broadband applications.

V. Conclusion

A 4×1 microstrip patch antenna array operating at $5.8\,\mathrm{GHz}$ was designed and matched to a 50 Ω SMA port with a feeding network. Performance was validated in Ansys HFSS, and all design specifications were met or exceeded. A stacked patch array configuration was then implemented to enhance the effective bandwidth. The measured bandwidth increased to 1.3 GHz—well above the predicted 165 MHz enhancement. This unusual bandwidth is attributed to stronger-than-designed coupling, combined with effects from higher-order patch radiation modes. Discrepancies between the simulated and measured radiation patterns were due to poor alignment tolerances in the stacked assembly. Future work could address tolerance issues with custom standoffs, explore lower-loss substrates to improve radiation efficiency, or experiment with mitered feeding-network corners to improve impedance matching.

References

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[3] R. Bancroft, *Microstrip and Printed Antenna Design*, 2nd ed. Raleigh, NC: SciTech Publishing, 2009.
[4] D. M. Pozar and S. M. Duffy, “A dual-band circularly polarized aperture-coupled stacked microstrip antenna for global positioning satellite,” *IEEE Trans. Antennas Propagat.*, vol. 45, no. 11, pp. 1618–1625, Nov. 1997.